

A Phase Transition on the Third Subset Diagonal of the Avoid-or-Compress Lemma

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Abstract

We determine the exact worst-case avoid-or-compress distance on the third subset diagonal $|S| - |A| = 3$ for synchronizing strongly connected automata. If $s = |S|$, then the tight value is

$$T_3(n, s) = \begin{cases} 1, & s = n, \\ 3, & s = n - 1, \\ n - s + 3, & s \leq n - 2. \end{cases}$$

The first two rows are forced by classical compression: a synchronizing automaton has a singular letter, and an $(n-1)$ -subset can be compressed within three steps. For every $s \leq n-2$ we construct an explicit binary, strongly connected, synchronizing automaton attaining the general avoid-or-compress upper bound $n - |A| = n - s + 3$. The proper-subset constructions split according to $L = n - |A|$ modulo three. Their lower bound is governed by a three-token corridor on a cyclic block together with a constant-size tail defect. Synchronization is proved symbolically using full cycles and local merges, or an explicit reset word. An independent verifier checks 1,680 constructions with $1 \leq |A| \leq 30$ and $5 \leq n - |A| \leq 60$, up to $n = 90$, and verifies the exact distance, strong connectivity, pair synchronizability, and the identities used in the synchronization proof. The result exhibits an exact one-step phase transition between the boundary $|S| = n - 1$ and the proper region $|S| \leq n - 2$.

1 Introduction

The avoid-or-compress lemma is a useful local tool in synchronizing automata: for nonempty $A \subsetneq S \subseteq Q$, it bounds the length required either to remove at least one state of A from the image of S or to compress S itself. In the synchronizing setting, the general upper bound is $n - |A|$ [1, 2]. The recent survey of open problems asks for parameter-dependent tight bounds and stresses that extremal examples should ideally remain synchronizing and strongly connected [2].

The second subset diagonal $|S| - |A| = 2$ admits exact tightness throughout the proper-subset region [4]. The third diagonal is the first place where a different phenomenon appears. The classical Frankl–Pin one-step subset-compression bound

$$\binom{n - |S| + 2}{2}$$

can be smaller than $n - |A|$. Exactly one boundary layer is affected: when $|S| = n - 1$, compression gives 3 whereas the avoid-or-compress lemma gives 4. One step farther inside, at $|S| = n - 2$, the two bounds both equal 5, and the general avoid-or-compress value becomes attainable again.

We prove that this comparison is not merely an upper-bound artifact: it gives the exact phase transition. The lower-bound constructions use a common cyclic architecture. All states of

A form a chain under one letter; the complement B is a cycle under the other. The three active B -states behave like tokens moving through a periodic coloring. Away from a constant-size defect near the end of the cycle, every use of the non-cyclic letter reconstructs the initial subset exactly. Thus success is delayed until the token window reaches the defect.

2 Preliminaries

Let $\mathcal{A} = (Q, \Sigma, \delta)$ be a DFA with $|Q| = n$. For $X \subseteq Q$ and $w \in \Sigma^*$, write Xw for the image of X .

Definition 1. For $\emptyset \neq A \subsetneq S \subseteq Q$, define

$$\tau_{\text{ac}}(\mathcal{A}; S, A) = \min\{|w| : A \not\subseteq Sw \text{ or } |Sw| < |S|\}.$$

On the third diagonal let

$$|S| - |A| = 3,$$

and write

$$r = |A|, \quad L = n - r = n - |A|.$$

Then $|S| = r + 3$ and $n - |S| = L - 3$.

Two general upper bounds are relevant. The avoid-or-compress lemma gives

$$\tau_{\text{ac}}(\mathcal{A}; S, A) \leq L, \tag{1}$$

while the Frankl–Pin subset-compression bound gives a compressing word of length at most

$$\binom{n - |S| + 2}{2} = \binom{L - 1}{2}. \tag{2}$$

The combinatorial extremal estimate underlying this compression theorem was proved by Frankl in response to a problem posed by Pin [3]; the resulting automata bound is commonly referred to as the Frankl–Pin or Pin–Frankl bound [5]. See also [1, 2].

3 Exact theorem and the boundary phase transition

Theorem 1 (Exact third diagonal). *Let $\mathcal{A}$ range over synchronizing strongly connected DFAs, and let $\emptyset \neq A \subsetneq S \subseteq Q$ satisfy $|S| - |A| = 3$. Put $s = |S|$. The tight worst-case value is*

$$T_3(n, s) = \begin{cases} 1, & s = n, \\ 3, & s = n - 1, \\ n - s + 3, & s \leq n - 2. \end{cases}$$

Moreover, every lower bound in the last two rows is attained by a binary, strongly connected, synchronizing automaton.

Upper bounds. If $s = n$, synchronization implies the existence of a singular letter, hence $T_3(n, n) \leq 1$; the empty word is not successful, so equality holds.

If $s = n - 1$, (2) gives a compressing word of length at most $\binom{3}{2} = 3$, hence $T_3(n, n - 1) \leq 3$.

If $s \leq n - 2$, the general avoid-or-compress bound gives $T_3(n, s) \leq n - s + 3$. The constructions below meet these bounds. $\square$

3.1 The $s = n - 1$ lower bound

Take the standard binary n -state Černý automaton with states $q_0, \dots, q_{n-1}$, where a is the n -cycle and b fixes all states except $q_{n-1}b = q_0$. Let

$$S = Q \setminus \{q_{n-1}\}, \quad A = \{q_2, \dots, q_{n-3}\}.$$

For $n \geq 5$, $|S| - |A| = 3$. No word of length at most two either compresses S or removes a state of A , whereas

$$Sa^3 = Q \setminus \{q_2\}$$

avoids q_2 . Thus $\tau_{ac} = 3$. The automaton is binary, strongly connected, and synchronizing.

4 Proper-subset constructions

We now assume $s \leq n - 2$, equivalently $L \geq 5$. Let

$$A = \{A_0, \dots, A_{r-1}\}, \quad B = \{B_0, \dots, B_{L-1}\},$$

$$Q = A \cup B, \quad S = A \cup \{B_0, B_1, B_2\}.$$

Letter b fixes A and cycles B :

$$A_i b = A_i, \quad B_j b = B_{j+1 \bmod L}.$$

On A , letter a is the chain

$$A_0 \rightarrow A_1 \rightarrow \dots \rightarrow A_{r-1} \rightarrow B_0.$$

The action of a on B depends on $L \bmod 3$.

4.1 $L \equiv 2 \pmod{3}$

For $0 \leq j < L - 2$, set

$$B_j a = \begin{cases} B_1, & j \equiv 0 \pmod{3}, \\ A_0, & j \equiv 1 \pmod{3}, \\ B_2, & j \equiv 2 \pmod{3}, \end{cases}$$

and at the tail set

$$B_{L-2}a = B_1, \quad B_{L-1}a = A_0.$$

A shortest successful word is

$$w_2 = b^{L-2}a^2. \tag{3}$$

4.2 $L \equiv 0 \pmod{3}$

For $0 \leq j < L - 1$, set

$$B_j a = \begin{cases} A_0, & j \equiv 0 \pmod{3}, \\ B_2, & j \equiv 1 \pmod{3}, \\ B_1, & j \equiv 2 \pmod{3}, \end{cases}$$

and set

$$B_{L-1}a = B_{L-1}.$$

A shortest successful word is

$$w_0 = b^{L-3}aba. \tag{4}$$

4.3 $L \equiv 1 \pmod{3}$

Here $L \geq 7$. For $0 \leq j < L - 1$, set

$$B_j a = \begin{cases} B_1, & j \equiv 0 \pmod{3}, \\ B_2, & j \equiv 1 \pmod{3}, \\ A_0, & j \equiv 2 \pmod{3}, \end{cases}$$

and define the single tail defect

$$B_{L-1} a = B_{L-3}.$$

A shortest successful word is

$$w_1 = b^{L-2} a^2. \tag{5}$$

5 Lower bound: three-token corridor and defect zone

The lower-bound proof is the same in spirit in all three residue classes. The A -chain contributes

$$A \setminus \{A_0\} \quad \text{and} \quad B_0$$

after applying a . Hence, for a safe subset of the form $A \cup T$ with $T \subseteq B$ and $|T| = 3$, the three B -images under a must provide A_0 together with two distinct B -states different from B_0 . In the regular part of each construction, every three consecutive B -states do exactly this.

Lemma 1 (Corridor reset). *Suppose the moving window*

$$W_k = \{B_k, B_{k+1}, B_{k+2}\}$$

does not intersect the residue-class tail defect. Then

$$(A \cup W_k) a = S.$$

Consequently, any occurrence of a in the regular corridor erases the preceding displacement and returns the active subset exactly to S .

Proof. In each residue class, the regular period has length three and the three images are precisely a permutation of $\{A_0, B_1, B_2\}$. The A -chain supplies $A_1, \dots, A_{r-1}, B_0$. Their union is S . $\square$

Thus a shortest successful word cannot use its first a while the three-token window is in the regular corridor: deleting the prefix through that a would produce a shorter successful word. The first a of a shortest witness must occur only after the window reaches the final constant-size defect zone.

Lemma 2 (Defect-zone check). *For each of the three residue classes, every word of length less than L whose first a occurs in the defect zone is safe. The words (3), (4), and (5), respectively, are successful and have length L .*

Proof. For compactness write

$$X(i, j, k) = A \cup \{B_i, B_j, B_k\},$$

with B -indices interpreted modulo L . Every displayed $X(i, j, k)$ is safe because it contains all of A and exactly three distinct B -states. Before the first occurrence of a a word is necessarily a power b^t . By Lemma 1, every value of t not listed below is in the regular corridor and the first a returns the image to S . It therefore suffices to inspect the final constant number of values of t and the suffixes that still fit inside a word of total length $< L$.

Case $L \equiv 2 \pmod{3}$. First assume $L \geq 8$. The only defect displacements relevant below length L are $t = L - 4, L - 3, L - 2$. For $t = L - 4$ the first a gives $X(0, 2, 4)$ and at most two further letters are available. Direct substitution gives

u	ε	a	b	aa	ab	ba	bb
$X(0, 2, 4)u$	$X(0, 2, 4)$	S	$X(1, 3, 5)$	S	$X(1, 2, 3)$	S	$X(2, 4, 6)$.

All seven images are safe. For $t = L - 3$ the first image is again $X(0, 2, 4)$ but only suffixes of length at most one are possible, so this is already covered. For $t = L - 2$ the first image is $X(0, 1, 4)$ and no further letter is allowed below length L . The remaining displacement $t = L - 1$ places the first a at length L itself. The exceptional value $L = 5$ has the same conclusion; wrap-around changes only

$$X(0, 2, 4)b = X(0, 1, 3), \quad X(0, 2, 4)ba = X(0, 1, 4), \quad X(0, 2, 4)bb = X(1, 2, 4),$$

and these images are still safe. Finally, from $X(0, 1, 4)$ a further a compresses, so $b^{L-2}a^2$ succeeds at length L .

Case $L \equiv 0 \pmod{3}$. The only relevant defect displacements are $t = L - 3$ and $t = L - 2$. In both cases the first a gives

$$X(0, 2, L - 1).$$

For $t = L - 3$ there is room for at most one further letter, and

$$X(0, 2, L - 1)a = X(0, 1, L - 1), \quad X(0, 2, L - 1)b = X(0, 1, 3),$$

both safe. For $t = L - 2$ no further letter is available below length L . The next displacement puts the first a at length L . Moreover,

$$X(0, 2, L - 1)b = X(0, 1, 3),$$

and a final a compresses this set, proving that $b^{L-3}aba$ succeeds at length L .

Case $L \equiv 1 \pmod{3}$. Here $L \geq 7$. The only relevant defect displacements are $t = L - 3$ and $t = L - 2$. At $t = L - 3$ the first a gives

$$X(0, 2, L - 3),$$

and the only possible one-letter continuations below length L satisfy

$$X(0, 2, L - 3)a = S, \quad X(0, 2, L - 3)b = X(1, 3, L - 2),$$

so both remain safe. At $t = L - 2$ the first a gives $X(0, 1, L - 3)$ and no further letter is available in a word shorter than L . A second a maps two of its three B -states to the same state and compresses, so $b^{L-2}a^2$ succeeds at length L .

These lists exhaust every location of the first a and every continuation compatible with total length $< L$. Hence every shorter word is safe in all three residue classes, while the displayed length- L words succeed. The verifier in Section 8 independently reproduces the same finite transitions over the full parameter grid. $\square$

Proposition 1 (Exact proper-subset distance). *For every $r \geq 1$ and $L \geq 5$, the corresponding construction satisfies*

$$\tau_{\text{ac}}(\mathcal{A}; S, A) = L = n - |A|.$$

Proof. Lemma 1 forces a shortest successful word into the tail defect, and Lemma 2 excludes success before length L while providing a successful word of length exactly L . $\square$

6 Strong connectivity

Proposition 2. *Every proper-subset construction is strongly connected.*

Proof. Every A_i reaches B_0 by powers of a , and b traverses the entire B -cycle. In every residue-class construction at least one B_j satisfies $B_j a = A_0$. Thus every state reaches A_0 , and A_0 reaches every state. $\square$

7 Synchronization

We use the following standard observation.

Lemma 3 (Cycle-and-merge). *Let C be a finite set on which a word α acts as a full cycle. If the transformation semigroup contains an idempotent that fixes all but one point of C and sends that point to an adjacent point on the α -cycle, then C is synchronizable. The same holds if an idempotent fixes all but the last two consecutive points and maps both of those points to their common predecessor.*

Proof. Conjugating the local merge by powers of the full cycle moves it to every cyclic position. Successively applying these conjugates reduces the image size until one state remains. $\square$

7.1 $L \equiv 1 \pmod{3}$

Let

$$C = A \cup \{B_0, B_1, B_2\}, \quad m = r + 3.$$

Directly,

$$Qa^2 = C,$$

and a acts on C as the full m -cycle

$$A_0 \rightarrow \cdots \rightarrow A_{r-1} \rightarrow B_0 \rightarrow B_1 \rightarrow B_2 \rightarrow A_0.$$

Set

$$f = b^{L-2}a^2, \quad \delta = fa^{m-2}, \quad e = \delta^2.$$

A direct substitution shows that e fixes every core position except the penultimate one, which it sends to the last. By Lemma 3, C and hence Q synchronize.

7.2 $L \equiv 2 \pmod{3}$

Again $Qa^2 = C = A \cup \{B_0, B_1, B_2\}$. Moreover,

$$Cb^{L-2}a^2 = D := A \cup \{B_0, B_1\},$$

and a is a full cycle on D , of size $m = r + 2$.

For $L \geq 8$, set

$$\gamma = b^2ab^{L-2}a^2, \quad \delta = \gamma a^{m-3}, \quad e = \delta^2.$$

Then e fixes positions $0, \dots, m-3$ and maps the last two positions to position $m-3$. Lemma 3 synchronizes D . For the exceptional case $L = 5$, the word $\gamma = b^2aba^2$ yields, after the same normalization, a standard adjacent idempotent merge.

7.3 $L \equiv 0 \pmod{3}$

Here an explicit reset word is available:

$$R = ab^{L-2}a(b^{L-1}a)^{r+1}ba. \quad (6)$$

After the prefix $ab^{L-2}a$, the image is

$$X_0 = A \cup \{B_0, B_2, B_{L-1}\}.$$

Let $T = b^{L-1}a$. For $0 \leq k < r$,

$$X_k = \{A_k, \dots, A_{r-1}\} \cup \{B_0, B_2, B_{L-1}\}$$

satisfies $X_k T = X_{k+1}$. After $r+1$ applications of T , the image is $\{B_2, B_{L-1}\}$, and

$$\{B_2, B_{L-1}\}ba = \{A_0\}.$$

Hence (6) is a reset word.

Combining these three cases with Proposition 2 completes the lower-bound construction required for Theorem 1.

8 Computational audit

An independent verifier constructs every residue-class family and checks:

- (i) exact breadth-first first-success distance L ;
- (ii) success of the closed-form length- L witness;
- (iii) strong connectivity;
- (iv) synchronizability via the exact unordered-pair criterion;
- (v) the symbolic core identities used in the synchronization proof.

For the grid

$$1 \leq r \leq 30, \quad 5 \leq L \leq 60,$$

there are 1,680 proper-subset constructions, up to $n = 90$. All 1,680 pass every check. The boundary $L = 3$ is the full-set case, and $L = 4$ is handled by the compression upper bound and the Černý witness.

Audit item	Result
Proper-subset parameter pairs	1,680
Exact first-success distance	1,680 / 1,680
Closed-form witness	1,680 / 1,680
Strong connectivity	1,680 / 1,680
Pair synchronizability	1,680 / 1,680
Proof-level core identity	1,680 / 1,680
Failures	0

9 Discussion

The third diagonal is the first diagonal on which the avoid-or-compress upper bound and the classical one-step compression bound cross. The exact theorem shows that the crossing produces a genuine extremal phase transition, not merely a proof artifact:

$$1, 3, 5, 6, 7, \dots$$

are the successive worst-case values as the codimension $n - |S|$ grows from 0 upward. In particular, the general value $n - |A|$ becomes sharp immediately once $|S| \leq n - 2$.

The next unresolved point is no longer the third diagonal itself but the interaction between diagonal index and codimension in the region where the Frankl–Pin bound competes closely with avoid-or-compress. The companion structural paper studies the concrete equality candidate $(d, k) = (3, 7)$, where both general bounds equal ten.

Data and code availability

The accompanying script `N018_THIRD_DIAGONAL_VERIFY_V3.py` reproduces the 1,680-case audit and the symbolic synchronization identities.

References

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